

Assignment Sheet: Project Summary

ECE 2031: Digital Design Laboratory

Length: 650 words maximum (~2 pages of text, not including figures).

There is no minimum. Include the required content as clearly and concisely as possible

Point Distribution: 150 points: introduction and conclusion scored individually; document body created collectively and scored for team

Due Date: Final Instructional Day at 8pm

Description of Project Summary:

The audience for this document is people who have been given your peripheral as part of a future project – possibly to extend it, or as an example of creating an SCOMP peripheral. In addition to needing to understand the basic functionality of your design, they need to know a bit about your design process so that they can understand why you made the choices you made.

Note that this is not a technical reference, application note, or datasheet – the intended audience is not someone directly using your peripheral who needs to know the exact function of everything. Although you *would* create that documentation for a full design project, you do *not* create it in 2031. This document is a project summary.

Document Content

There is a fundamental concept critical to this document: **you will not be able to fit a description of every aspect of the project.** The intention is for you to select the most important and appropriate information within the allowed length. Approach the design of this document from the very beginning with the understanding that you cannot include everything about your project and that you must, instead, choose only the most important aspects.

Images do not contribute to the word count, so using them will allow you to convey a lot more information than just text. The word count also does *not* include:

- The title page.
- Simple labels within figures (e.g. things like “SCOMP” or “Peripheral” labeling a box in a block diagram)
 - Annotations that *add* information, as opposed to simply labeling or disambiguating, are included. You can’t skirt the word limit by putting text in images.
 - Figure captions are included. This is a document that benefits greatly from figures being captioned, and those captions contain information about interpreting the figure.
- Table labels and data (e.g. things like “I/O Address” in a column header and “0xA0” in a cell).
 - Something like a description of behavior within a table cell would be included. You can’t skirt the limit by explaining your peripheral’s functionality in a table. You can do that in a table if desired (and it could be an excellent way to summarize your peripheral’s I/O registers); it just counts toward the word limit.

Based on questions from previous semesters, here are a couple specific things that do count toward the word limit:

- Code (ASM or VHDL). It is unlikely that code is useful for this document's user anyway, but you can include code if you think it is worth the words.
- Information in appendices, using the same exclusions listed previously. You can include appendices if that is the best place to put information, but you can't use appendices as a dumping ground for lots of information.

It's not possible to list every possible scenario here where you might use words, and you shouldn't fixate on it. In general, if text contributes significant information then it counts, because the purpose of this document is for you to critically consider what is most important about your project for the intended audience and how to concisely present that information.

Organize the document into the following four sections:

Introduction

- Describe the document's contents and purpose.
- Introduce the capabilities and behavior of your device.
- Briefly discuss your approach to the project. Were there features or attributes of the peripheral that you prioritized? Was there anything unique or important about how you organized your team or managed your time?
- Summarize how well the project was accomplished (did your final product meet the required specifications and your intentions outlined in your original proposal? If not, why not?).

These items are not necessarily in the order that should appear in your intro. Ensure that the entire introduction makes sense and reads as a cohesive section.

Helpful Hints: The introduction works to lay the foundation of the document, preparing the reader for the details that will be presented in the rest of the summary. The introduction is not a teaser – its purpose is not to entice the reader to continue, but to inform them about the document content so that they know whether or not this document fits their needs, and if it does, to give them a mental framework that they can fill in with details as they continue reading.

Device Functionality

- Describe how a user (an SCOMP programmer) uses your peripheral.
 - Note that this is not the application that you created to *demonstrate* your peripheral.
- If applicable, include any special features or explain any special situations that apply to your peripheral.
- Include relevant graphs, tables, and figures to make the information easier to understand.

This is where it is particularly important to remember that this document is not a complete summary of everything that you did and everything that your peripheral does. Quickly summarize the behavior, focusing on the most important aspects relevant to the audience, and don't get bogged down in low-level details.

Design Decisions and Implementation

- Give a description of your team's choices within the design space and why you made those choices.
- Explain how those choices affected your hardware and your approach to the project.
- Discuss any significant modifications that were made during the design process (you don't need to present every change you made, just the most important ones).

Conclusions

- Provide your overall conclusions on your project, relating to the original purpose of the project.
- Make any recommendations you feel appropriate (recommendations related to YOUR work and how future engineers might use it; this report is not the place to discuss the projects in general).
- In retrospect, how could you have optimized your design or improved your process? What would you do differently if you had to revisit this project in the future?

Helpful Hints: Re-stating or summarizing information from previous section is not a conclusion. A conclusion in this context is something that you can conclude based on what you have discussed. The conclusions section is a place to discuss, analyze, critique, and evaluate the overall design and your path through the project. We will be grading your ability to think critically about your project, including the technical aspects and the design process.

Expected Collaboration:

The introduction and conclusion should be your own individual work. In the introduction, you should explain the project and your approach in your own words, as relevant to this document. The conclusion should contain your own personal analysis of the results and your own recommendations for future engineers.

However, your team worked on this project together and created a single design, so the device functionality and design decisions will include the same information. Work with your team to create those sections, and use the text and figures of those sections for all submissions.

Our recommendation is that you work as a team to create a draft of those sections first, iterate a few times to converge on a final version of the collaborative sections, then base your personal introduction and conclusion on that. That is usually the best way to create any piece of communication – you don't know specifically what you are introducing or concluding until the body information is complete.

Your document as a whole must be cohesive! You will lose points if your introduction does not fit with the body of the document, or if your conclusions do not follow from the presented information.